

# Fragility Functions

**Table 1.** Fragility function parameters for RC buildings in L'Aquila for the respective  $T_1$  using CS set of records.

| Taxonomy                                 | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|------------------------------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                                          |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1<br>$T = 0.14\text{ s}$ | $Sa(T)$ [g]        | 0.51   | 0.30    | 1.09   | 0.48    | 1.89   | 0.67    | 4.76   | 1.02    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.42   | 0.28    | 0.86   | 0.38    | 1.30   | 0.47    | 2.63   | 0.73    |
|                                          | $FIV3(T)$ [cm/s]   | 9.12   | 0.80    | 18.04  | 0.53    | 28.34  | 0.24    | 45.17  | 0.35    |
| CR-LFINF-CDM-7-H1<br>$T = 0.14\text{ s}$ | $Sa(T)$ [g]        | 0.50   | 0.32    | 1.09   | 0.47    | 1.88   | 0.65    | 4.54   | 0.95    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.41   | 0.27    | 0.85   | 0.40    | 1.32   | 0.49    | 2.66   | 0.70    |
|                                          | $FIV3(T)$ [cm/s]   | 8.73   | 0.81    | 18.06  | 0.53    | 28.83  | 0.35    | 41.34  | 0.40    |
| CR-LFINF-CDL-7-H2<br>$T = 0.25\text{ s}$ | $Sa(T)$ [g]        | 0.34   | 0.33    | 0.71   | 0.40    | 1.23   | 0.55    | 2.29   | 0.79    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.28   | 0.22    | 0.52   | 0.28    | 0.88   | 0.52    | 1.68   | 0.62    |
|                                          | $FIV3(T)$ [cm/s]   | 10.56  | 0.69    | 22.58  | 0.53    | 36.78  | 0.33    | 55.92  | 0.27    |
| CR-LFINF-CDM-7-H2<br>$T = 0.27\text{ s}$ | $Sa(T)$ [g]        | 0.28   | 0.34    | 0.54   | 0.35    | 1.06   | 0.54    | 2.03   | 0.70    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.23   | 0.22    | 0.48   | 0.31    | 0.77   | 0.44    | 1.39   | 0.57    |
|                                          | $FIV3(T)$ [cm/s]   | 9.56   | 0.70    | 22.69  | 0.53    | 34.23  | 0.36    | 53.01  | 0.25    |
| CR-LFINF-CDL-7-H3<br>$T = 0.35\text{ s}$ | $Sa(T)$ [g]        | 0.24   | 0.30    | 0.55   | 0.45    | 0.95   | 0.50    | 1.98   | 0.80    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.22   | 0.22    | 0.42   | 0.29    | 0.81   | 0.55    | 1.34   | 0.63    |
|                                          | $FIV3(T)$ [cm/s]   | 12.31  | 0.58    | 26.07  | 0.50    | 47.16  | 0.29    | 59.28  | 0.25    |
| CR-LFINF-CDM-7-H3<br>$T = 0.37\text{ s}$ | $Sa(T)$ [g]        | 0.21   | 0.24    | 0.44   | 0.35    | 0.83   | 0.60    | 1.40   | 0.65    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.18   | 0.27    | 0.37   | 0.28    | 0.61   | 0.49    | 0.91   | 0.49    |
|                                          | $FIV3(T)$ [cm/s]   | 10.99  | 0.53    | 24.50  | 0.38    | 39.02  | 0.25    | 58.36  | 0.21    |
| CR-LFINF-CDL-7-H4<br>$T = 0.44\text{ s}$ | $Sa(T)$ [g]        | 0.20   | 0.29    | 0.41   | 0.35    | 0.68   | 0.40    | 1.41   | 0.59    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.16   | 0.25    | 0.33   | 0.21    | 0.51   | 0.27    | 1.02   | 0.54    |
|                                          | $FIV3(T)$ [cm/s]   | 11.48  | 0.53    | 25.49  | 0.42    | 43.42  | 0.41    | 51.05  | 0.35    |
| CR-LFINF-CDM-7-H4<br>$T = 0.45\text{ s}$ | $Sa(T)$ [g]        | 0.19   | 0.27    | 0.41   | 0.35    | 0.72   | 0.45    | 1.26   | 0.67    |
|                                          | $Sa_{avg2}(T)$ [g] | 0.16   | 0.19    | 0.33   | 0.26    | 0.58   | 0.45    | 0.98   | 0.51    |
|                                          | $FIV3(T)$ [cm/s]   | 11.30  | 0.54    | 29.24  | 0.35    | 47.51  | 0.25    | 67.17  | 0.19    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

CS: conditional spectrum.

**Table 2.** Fragility function parameters for RC buildings in L'Aquila for T=0.14 using CS set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.51   | 0.30    | 1.09   | 0.48    | 1.89   | 0.67    | 4.76   | 1.02    |
|                   | $Sa_{avg2}(T)$ [g] | 0.42   | 0.28    | 0.86   | 0.38    | 1.30   | 0.47    | 2.63   | 0.73    |
|                   | $FIV3(T)$ [cm/s]   | 9.12   | 0.80    | 18.04  | 0.53    | 28.34  | 0.24    | 45.17  | 0.35    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.50   | 0.32    | 1.09   | 0.47    | 1.88   | 0.65    | 4.54   | 0.95    |
|                   | $Sa_{avg2}(T)$ [g] | 0.41   | 0.27    | 0.85   | 0.40    | 1.32   | 0.49    | 2.66   | 0.70    |
|                   | $FIV3(T)$ [cm/s]   | 8.73   | 0.81    | 18.06  | 0.53    | 28.83  | 0.35    | 41.34  | 0.40    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.39   | 0.55    | 0.77   | 0.64    | 1.60   | 0.80    | 3.41   | 0.94    |
|                   | $Sa_{avg2}(T)$ [g] | 0.27   | 0.56    | 0.61   | 0.54    | 1.01   | 0.56    | 1.99   | 0.73    |
|                   | $FIV3(T)$ [cm/s]   | 6.00   | 0.65    | 12.43  | 0.46    | 19.54  | 0.34    | 34.29  | 0.21    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.35   | 0.45    | 0.68   | 0.67    | 1.38   | 0.83    | 2.93   | 0.95    |
|                   | $Sa_{avg2}(T)$ [g] | 0.26   | 0.53    | 0.52   | 0.55    | 0.89   | 0.60    | 1.79   | 0.79    |
|                   | $FIV3(T)$ [cm/s]   | 5.36   | 0.64    | 10.91  | 0.43    | 18.53  | 0.26    | 30.88  | 0.25    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.32   | 0.67    | 0.77   | 0.75    | 1.66   | 0.90    | 3.55   | 1.15    |
|                   | $Sa_{avg2}(T)$ [g] | 0.26   | 0.44    | 0.58   | 0.54    | 1.12   | 0.65    | 2.13   | 0.82    |
|                   | $FIV3(T)$ [cm/s]   | 4.97   | 0.55    | 11.54  | 0.35    | 19.90  | 0.22    | 33.57  | 0.38    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.30   | 0.67    | 0.64   | 0.80    | 1.29   | 0.84    | 2.53   | 1.12    |
|                   | $Sa_{avg2}(T)$ [g] | 0.23   | 0.54    | 0.51   | 0.56    | 0.94   | 0.68    | 1.52   | 0.86    |
|                   | $FIV3(T)$ [cm/s]   | 4.38   | 0.57    | 10.47  | 0.29    | 17.20  | 0.18    | 26.72  | 0.29    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.30   | 0.63    | 0.64   | 0.71    | 1.25   | 0.86    | 2.93   | 1.11    |
|                   | $Sa_{avg2}(T)$ [g] | 0.24   | 0.50    | 0.53   | 0.55    | 0.85   | 0.68    | 2.01   | 0.96    |
|                   | $FIV3(T)$ [cm/s]   | 4.82   | 0.47    | 10.83  | 0.33    | 18.31  | 0.47    | 33.48  | 0.43    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.29   | 0.66    | 0.66   | 0.77    | 1.40   | 0.87    | 2.89   | 1.16    |
|                   | $Sa_{avg2}(T)$ [g] | 0.24   | 0.56    | 0.54   | 0.56    | 0.98   | 0.70    | 1.62   | 0.87    |
|                   | $FIV3(T)$ [cm/s]   | 4.60   | 0.36    | 10.98  | 0.30    | 18.04  | 0.21    | 26.54  | 0.26    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 3.** Fragility function parameters for RC buildings in L'Aquila for  $T=0.33$  using CS set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.41   | 0.33    | 0.88   | 0.24    | 1.28   | 0.28    | 2.65   | 0.58    |
|                   | $Sa_{avg2}(T)$ [g] | 0.37   | 0.31    | 0.70   | 0.20    | 0.99   | 0.27    | 1.84   | 0.49    |
|                   | $FIV3(T)$ [cm/s]   | 21.19  | 0.90    | 47.91  | 0.64    | 60.24  | 0.51    | 101.01 | 0.28    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.42   | 0.34    | 0.86   | 0.26    | 1.31   | 0.29    | 2.59   | 0.46    |
|                   | $Sa_{avg2}(T)$ [g] | 0.35   | 0.33    | 0.70   | 0.20    | 1.00   | 0.24    | 1.86   | 0.37    |
|                   | $FIV3(T)$ [cm/s]   | 32.12  | 0.79    | 47.85  | 0.65    | 61.33  | 0.48    | 105.14 | 0.24    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.30   | 0.08    | 0.60   | 0.27    | 1.03   | 0.36    | 1.89   | 0.66    |
|                   | $Sa_{avg2}(T)$ [g] | 0.25   | 0.26    | 0.47   | 0.18    | 0.83   | 0.37    | 1.43   | 0.56    |
|                   | $FIV3(T)$ [cm/s]   | 23.23  | 0.57    | 29.39  | 0.49    | 43.88  | 0.34    | 75.91  | 0.24    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.28   | 0.07    | 0.53   | 0.27    | 0.90   | 0.43    | 1.72   | 0.63    |
|                   | $Sa_{avg2}(T)$ [g] | 0.21   | 0.26    | 0.42   | 0.15    | 0.72   | 0.44    | 1.17   | 0.48    |
|                   | $FIV3(T)$ [cm/s]   | 22.21  | 0.55    | 24.85  | 0.51    | 36.25  | 0.33    | 68.01  | 0.31    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.25   | 0.37    | 0.56   | 0.36    | 1.10   | 0.50    | 2.08   | 0.70    |
|                   | $Sa_{avg2}(T)$ [g] | 0.20   | 0.17    | 0.45   | 0.31    | 0.84   | 0.51    | 1.49   | 0.64    |
|                   | $FIV3(T)$ [cm/s]   | 20.12  | 0.46    | 24.93  | 0.37    | 42.54  | 0.25    | 61.94  | 0.17    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.23   | 0.38    | 0.47   | 0.46    | 0.88   | 0.69    | 1.51   | 0.74    |
|                   | $Sa_{avg2}(T)$ [g] | 0.18   | 0.15    | 0.40   | 0.33    | 0.71   | 0.51    | 1.08   | 0.50    |
|                   | $FIV3(T)$ [cm/s]   | 17.53  | 0.43    | 21.04  | 0.34    | 34.98  | 0.26    | 57.75  | 0.19    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.23   | 0.39    | 0.51   | 0.47    | 0.83   | 0.58    | 2.03   | 0.78    |
|                   | $Sa_{avg2}(T)$ [g] | 0.18   | 0.29    | 0.37   | 0.40    | 0.58   | 0.36    | 1.28   | 0.68    |
|                   | $FIV3(T)$ [cm/s]   | 17.83  | 0.50    | 22.84  | 0.39    | 37.29  | 0.45    | 67.22  | 0.54    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.23   | 0.38    | 0.51   | 0.49    | 0.86   | 0.61    | 1.79   | 0.80    |
|                   | $Sa_{avg2}(T)$ [g] | 0.18   | 0.35    | 0.38   | 0.45    | 0.66   | 0.48    | 1.21   | 0.53    |
|                   | $FIV3(T)$ [cm/s]   | 18.22  | 0.32    | 22.61  | 0.33    | 37.28  | 0.20    | 60.24  | 0.19    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 4.** Fragility function parameters for RC buildings in L'Aquila for  $T=0.45$  using CS set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.34   | 0.52    | 0.71   | 0.35    | 1.12   | 0.24    | 1.78   | 0.34    |
|                   | $Sa_{avg2}(T)$ [g] | 0.31   | 0.37    | 0.61   | 0.30    | 0.89   | 0.15    | 1.33   | 0.28    |
|                   | $FIV3(T)$ [cm/s]   | 21.57  | 0.96    | 48.96  | 0.70    | 74.95  | 0.44    | 118.67 | 0.31    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.34   | 0.53    | 0.71   | 0.36    | 1.10   | 0.24    | 1.94   | 0.37    |
|                   | $Sa_{avg2}(T)$ [g] | 0.31   | 0.36    | 0.61   | 0.30    | 0.90   | 0.15    | 1.41   | 0.23    |
|                   | $FIV3(T)$ [cm/s]   | 21.57  | 0.96    | 48.67  | 0.72    | 77.20  | 0.47    | 129.06 | 0.31    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.24   | 0.34    | 0.48   | 0.15    | 0.79   | 0.34    | 1.35   | 0.39    |
|                   | $Sa_{avg2}(T)$ [g] | 0.21   | 0.32    | 0.40   | 0.16    | 0.68   | 0.23    | 1.07   | 0.34    |
|                   | $FIV3(T)$ [cm/s]   | 14.80  | 0.72    | 33.61  | 0.57    | 52.08  | 0.38    | 80.95  | 0.31    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.20   | 0.27    | 0.43   | 0.18    | 0.69   | 0.36    | 1.29   | 0.43    |
|                   | $Sa_{avg2}(T)$ [g] | 0.19   | 0.30    | 0.36   | 0.17    | 0.61   | 0.23    | 0.97   | 0.33    |
|                   | $FIV3(T)$ [cm/s]   | 13.34  | 0.69    | 30.62  | 0.54    | 46.29  | 0.41    | 80.18  | 0.25    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.20   | 0.14    | 0.47   | 0.23    | 0.83   | 0.52    | 1.54   | 0.61    |
|                   | $Sa_{avg2}(T)$ [g] | 0.17   | 0.33    | 0.38   | 0.19    | 0.69   | 0.32    | 1.12   | 0.49    |
|                   | $FIV3(T)$ [cm/s]   | 11.45  | 0.62    | 31.95  | 0.44    | 54.37  | 0.30    | 76.64  | 0.21    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.19   | 0.17    | 0.40   | 0.30    | 0.68   | 0.42    | 1.10   | 0.59    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.29    | 0.33   | 0.18    | 0.54   | 0.38    | 0.88   | 0.45    |
|                   | $FIV3(T)$ [cm/s]   | 11.19  | 0.61    | 26.67  | 0.40    | 43.93  | 0.30    | 65.42  | 0.17    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.19   | 0.27    | 0.41   | 0.23    | 0.70   | 0.52    | 1.50   | 0.63    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.20    | 0.34   | 0.22    | 0.56   | 0.36    | 1.06   | 0.58    |
|                   | $FIV3(T)$ [cm/s]   | 11.41  | 0.60    | 27.68  | 0.46    | 45.07  | 0.43    | 85.02  | 0.41    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.19   | 0.27    | 0.41   | 0.35    | 0.72   | 0.45    | 1.26   | 0.67    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.19    | 0.33   | 0.26    | 0.58   | 0.45    | 0.98   | 0.51    |
|                   | $FIV3(T)$ [cm/s]   | 11.30  | 0.54    | 29.24  | 0.35    | 47.51  | 0.25    | 68.97  | 0.19    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 5.** Fragility function parameters for RC buildings in L'Aquila for T=0.7 using CS set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.28   | 0.73    | 0.55   | 0.50    | 0.79   | 0.35    | 1.28   | 0.21    |
|                   | $Sa_{avg2}(T)$ [g] | 0.21   | 0.63    | 0.46   | 0.35    | 0.67   | 0.22    | 0.98   | 0.20    |
|                   | $FIV3(T)$ [cm/s]   | 28.75  | 1.25    | 79.89  | 0.98    | 116.94 | 0.71    | 170.33 | 0.37    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.26   | 0.73    | 0.55   | 0.50    | 0.82   | 0.32    | 1.35   | 0.18    |
|                   | $Sa_{avg2}(T)$ [g] | 0.20   | 0.60    | 0.46   | 0.36    | 0.68   | 0.26    | 1.00   | 0.30    |
|                   | $FIV3(T)$ [cm/s]   | 27.86  | 1.18    | 80.60  | 0.98    | 112.07 | 0.76    | 181.32 | 0.36    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.16   | 0.55    | 0.33   | 0.36    | 0.58   | 0.24    | 0.93   | 0.32    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.39    | 0.29   | 0.32    | 0.46   | 0.20    | 0.75   | 0.24    |
|                   | $FIV3(T)$ [cm/s]   | 17.26  | 0.83    | 43.31  | 0.78    | 67.92  | 0.56    | 105.99 | 0.37    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.14   | 0.53    | 0.29   | 0.36    | 0.52   | 0.25    | 0.86   | 0.32    |
|                   | $Sa_{avg2}(T)$ [g] | 0.12   | 0.41    | 0.26   | 0.32    | 0.41   | 0.22    | 0.66   | 0.20    |
|                   | $FIV3(T)$ [cm/s]   | 15.77  | 0.75    | 38.70  | 0.71    | 59.18  | 0.55    | 94.23  | 0.42    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.15   | 0.38    | 0.32   | 0.21    | 0.57   | 0.33    | 0.92   | 0.55    |
|                   | $Sa_{avg2}(T)$ [g] | 0.14   | 0.34    | 0.28   | 0.20    | 0.47   | 0.20    | 0.78   | 0.44    |
|                   | $FIV3(T)$ [cm/s]   | 16.78  | 0.54    | 44.40  | 0.55    | 65.80  | 0.41    | 118.83 | 0.52    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.13   | 0.37    | 0.27   | 0.22    | 0.43   | 0.24    | 0.73   | 0.42    |
|                   | $Sa_{avg2}(T)$ [g] | 0.12   | 0.33    | 0.23   | 0.23    | 0.40   | 0.27    | 0.59   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 14.48  | 0.59    | 35.59  | 0.49    | 52.94  | 0.40    | 82.99  | 0.33    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.13   | 0.35    | 0.28   | 0.28    | 0.41   | 0.24    | 0.73   | 0.31    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.27    | 0.24   | 0.26    | 0.37   | 0.24    | 0.70   | 0.47    |
|                   | $FIV3(T)$ [cm/s]   | 14.03  | 0.64    | 34.62  | 0.55    | 58.22  | 0.59    | 118.61 | 0.49    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.12   | 0.27    | 0.29   | 0.21    | 0.49   | 0.31    | 0.77   | 0.47    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.27    | 0.26   | 0.16    | 0.40   | 0.20    | 0.64   | 0.32    |
|                   | $FIV3(T)$ [cm/s]   | 14.15  | 0.58    | 36.63  | 0.53    | 56.51  | 0.37    | 86.03  | 0.33    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 6.** Fragility function parameters for RC buildings in L'Aquila for T=1.5 using CS set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.11   | 1.07    | 0.25   | 1.01    | 0.39   | 0.69    | 0.62   | 0.45    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.82    | 0.24   | 0.73    | 0.31   | 0.52    | 0.48   | 0.33    |
|                   | $FIV3(T)$ [cm/s]   | 38.65  | 1.14    | 87.36  | 0.90    | 124.07 | 0.79    | 194.55 | 0.55    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.10   | 1.03    | 0.26   | 1.00    | 0.40   | 0.69    | 0.68   | 0.48    |
|                   | $Sa_{avg2}(T)$ [g] | 0.10   | 0.82    | 0.24   | 0.73    | 0.34   | 0.49    | 0.52   | 0.32    |
|                   | $FIV3(T)$ [cm/s]   | 38.16  | 1.16    | 89.89  | 0.92    | 127.54 | 0.79    | 195.87 | 0.50    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.09   | 0.80    | 0.14   | 0.69    | 0.24   | 0.46    | 0.41   | 0.43    |
|                   | $Sa_{avg2}(T)$ [g] | 0.06   | 0.75    | 0.13   | 0.50    | 0.20   | 0.46    | 0.36   | 0.37    |
|                   | $FIV3(T)$ [cm/s]   | 21.48  | 0.93    | 50.06  | 0.72    | 78.87  | 0.54    | 127.01 | 0.38    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.05   | 0.91    | 0.13   | 0.66    | 0.20   | 0.47    | 0.33   | 0.30    |
|                   | $Sa_{avg2}(T)$ [g] | 0.07   | 0.66    | 0.12   | 0.46    | 0.18   | 0.43    | 0.31   | 0.26    |
|                   | $FIV3(T)$ [cm/s]   | 19.65  | 0.81    | 42.01  | 0.74    | 64.92  | 0.55    | 109.95 | 0.40    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.04   | 0.75    | 0.13   | 0.56    | 0.23   | 0.33    | 0.38   | 0.48    |
|                   | $Sa_{avg2}(T)$ [g] | 0.04   | 0.54    | 0.12   | 0.49    | 0.21   | 0.44    | 0.34   | 0.43    |
|                   | $FIV3(T)$ [cm/s]   | 16.60  | 0.72    | 44.57  | 0.69    | 75.05  | 0.47    | 127.63 | 0.46    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.04   | 0.73    | 0.11   | 0.62    | 0.18   | 0.43    | 0.28   | 0.21    |
|                   | $Sa_{avg2}(T)$ [g] | 0.04   | 0.52    | 0.11   | 0.48    | 0.16   | 0.45    | 0.26   | 0.23    |
|                   | $FIV3(T)$ [cm/s]   | 15.20  | 0.70    | 38.49  | 0.67    | 65.36  | 0.44    | 94.47  | 0.31    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.04   | 0.70    | 0.11   | 0.69    | 0.18   | 0.62    | 0.36   | 0.50    |
|                   | $Sa_{avg2}(T)$ [g] | 0.04   | 0.54    | 0.11   | 0.54    | 0.17   | 0.59    | 0.31   | 0.55    |
|                   | $FIV3(T)$ [cm/s]   | 14.02  | 0.74    | 36.59  | 0.80    | 64.75  | 0.62    | 111.51 | 0.52    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.04   | 0.58    | 0.11   | 0.54    | 0.19   | 0.39    | 0.29   | 0.23    |
|                   | $Sa_{avg2}(T)$ [g] | 0.06   | 0.50    | 0.11   | 0.49    | 0.17   | 0.34    | 0.27   | 0.23    |
|                   | $FIV3(T)$ [cm/s]   | 14.42  | 0.74    | 37.60  | 0.61    | 67.11  | 0.45    | 94.10  | 0.34    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

CS: conditional spectrum.

**Table 7.** Fragility function parameters for RC buildings in L'Aquila for  $PGA$  using using CS set of records.

| Taxonomy          | IM        | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|-----------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |           | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $PGA$ [g] | 0.23   | 0.33    | 0.46   | 0.36    | 0.72   | 0.40    | 1.20   | 0.49    |
| CR-LFINF-CDM-7-H1 | $PGA$ [g] | 0.22   | 0.33    | 0.45   | 0.36    | 0.79   | 0.37    | 1.22   | 0.49    |
| CR-LFINF-CDL-7-H2 | $PGA$ [g] | 0.16   | 0.41    | 0.32   | 0.40    | 0.57   | 0.51    | 0.98   | 0.60    |
| CR-LFINF-CDM-7-H2 | $PGA$ [g] | 0.15   | 0.46    | 0.28   | 0.41    | 0.51   | 0.51    | 0.89   | 0.56    |
| CR-LFINF-CDL-7-H3 | $PGA$ [g] | 0.15   | 0.45    | 0.32   | 0.52    | 0.57   | 0.63    | 1.05   | 0.71    |
| CR-LFINF-CDM-7-H3 | $PGA$ [g] | 0.13   | 0.39    | 0.28   | 0.50    | 0.44   | 0.62    | 0.83   | 0.83    |
| CR-LFINF-CDL-7-H4 | $PGA$ [g] | 0.14   | 0.42    | 0.27   | 0.48    | 0.45   | 0.60    | 1.01   | 0.81    |
| CR-LFINF-CDM-7-H4 | $PGA$ [g] | 0.14   | 0.47    | 0.30   | 0.53    | 0.50   | 0.64    | 0.90   | 0.80    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

CS: conditional spectrum.

**Table 8.** Fragility function parameters for RC buildings in L'Aquila for the respective  $T_1$  using FEMAP695 set of records.

| Taxonomy                                 | IM                        | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|------------------------------------------|---------------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                                          |                           | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1<br>$T = 0.14\text{ s}$ | $Sa(T)$ [g]               | 0.44   | 0.26    | 0.72   | 0.34    | 0.93   | 0.38    | 1.40   | 0.41    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.44   | 0.06    | 0.69   | 0.19    | 0.88   | 0.24    | 1.20   | 0.29    |
|                                          | $FIV3(T)$ [cm/s]          | 36.64  | 0.23    | 62.39  | 0.12    | 76.53  | 0.05    | 108.11 | 0.22    |
| CR-LFINF-CDM-7-H1<br>$T = 0.14\text{ s}$ | $Sa(T)$ [g]               | 0.42   | 0.25    | 0.72   | 0.30    | 1.00   | 0.36    | 1.47   | 0.43    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.43   | 0.06    | 0.69   | 0.21    | 0.89   | 0.28    | 1.25   | 0.32    |
|                                          | $FIV3(T)$ [cm/s]          | 36.64  | 0.23    | 62.39  | 0.12    | 79.77  | 0.06    | 113.68 | 0.08    |
| CR-LFINF-CDL-7-H2<br>$T = 0.25\text{ s}$ | $Sa(T)$ [g]               | 0.29   | 0.20    | 0.53   | 0.28    | 0.70   | 0.34    | 1.16   | 0.53    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.27   | 0.08    | 0.49   | 0.17    | 0.67   | 0.21    | 1.04   | 0.34    |
|                                          | $FIV3(T)$ [cm/s]          | 28.90  | 0.20    | 53.02  | 0.19    | 79.25  | 0.12    | 119.51 | 0.26    |
| CR-LFINF-CDM-7-H2<br>$T = 0.27\text{ s}$ | $Sa(T)$ [g]               | 0.25   | 0.22    | 0.45   | 0.28    | 0.63   | 0.28    | 1.00   | 0.44    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.23   | 0.22    | 0.40   | 0.19    | 0.60   | 0.25    | 0.85   | 0.32    |
|                                          | $FIV3(T)$ [cm/s]          | 27.08  | 0.23    | 47.21  | 0.14    | 69.79  | 0.15    | 105.81 | 0.18    |
| CR-LFINF-CDL-7-H3<br>$T = 0.35\text{ s}$ | $Sa(T)$ [g]               | 0.20   | 0.34    | 0.43   | 0.27    | 0.69   | 0.38    | 1.12   | 0.57    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.19   | 0.18    | 0.39   | 0.21    | 0.63   | 0.23    | 1.07   | 0.54    |
|                                          | $FIV3(T)$ [cm/s]          | 22.55  | 0.26    | 50.08  | 0.16    | 81.84  | 0.13    | 128.26 | 0.31    |
| CR-LFINF-CDM-7-H3<br>$T = 0.37\text{ s}$ | $Sa(T)$ [g]               | 0.18   | 0.30    | 0.36   | 0.24    | 0.57   | 0.44    | 0.86   | 0.39    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.15   | 0.08    | 0.34   | 0.06    | 0.51   | 0.29    | 0.77   | 0.32    |
|                                          | $FIV3(T)$ [cm/s]          | 19.27  | 0.27    | 44.09  | 0.07    | 67.40  | 0.19    | 99.14  | 0.05    |
| CR-LFINF-CDL-7-H4<br>$T = 0.44\text{ s}$ | $Sa(T)$ [g]               | 0.17   | 0.26    | 0.35   | 0.30    | 0.54   | 0.29    | 0.99   | 0.51    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.16   | 0.22    | 0.32   | 0.19    | 0.49   | 0.20    | 0.89   | 0.46    |
|                                          | $FIV3(T)$ [cm/s]          | 20.89  | 0.24    | 44.61  | 0.20    | 69.03  | 0.20    | 115.06 | 0.18    |
| CR-LFINF-CDM-7-H4<br>$T = 0.45\text{ s}$ | $Sa(T)$ [g]               | 0.18   | 0.28    | 0.36   | 0.25    | 0.61   | 0.31    | 0.87   | 0.45    |
|                                          | $Sa_{\text{avg2}}(T)$ [g] | 0.16   | 0.19    | 0.34   | 0.05    | 0.55   | 0.28    | 0.77   | 0.31    |
|                                          | $FIV3(T)$ [cm/s]          | 20.53  | 0.26    | 47.15  | 0.18    | 75.86  | 0.06    | 102.08 | 0.22    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

**Table 9.** Fragility function parameters for RC buildings in L’Aquila for T=0.14 using FEMAP695 set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.44   | 0.26    | 0.72   | 0.34    | 0.93   | 0.38    | 1.40   | 0.41    |
|                   | $Sa_{avg2}(T)$ [g] | 0.44   | 0.06    | 0.69   | 0.19    | 0.88   | 0.24    | 1.20   | 0.29    |
|                   | $FIV3(T)$ [cm/s]   | 36.64  | 0.23    | 62.39  | 0.12    | 76.62  | 0.05    | 108.11 | 0.22    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.42   | 0.25    | 0.72   | 0.30    | 1.00   | 0.36    | 1.47   | 0.43    |
|                   | $Sa_{avg2}(T)$ [g] | 0.43   | 0.06    | 0.69   | 0.21    | 0.89   | 0.28    | 1.25   | 0.32    |
|                   | $FIV3(T)$ [cm/s]   | 36.64  | 0.23    | 62.39  | 0.12    | 79.60  | 0.06    | 116.31 | 0.01    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.24   | 0.37    | 0.46   | 0.40    | 0.65   | 0.39    | 1.02   | 0.43    |
|                   | $Sa_{avg2}(T)$ [g] | 0.24   | 0.21    | 0.42   | 0.31    | 0.59   | 0.30    | 0.87   | 0.30    |
|                   | $FIV3(T)$ [cm/s]   | 21.52  | 0.18    | 40.92  | 0.16    | 55.84  | 0.17    | 83.97  | 0.30    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.22   | 0.43    | 0.39   | 0.44    | 0.59   | 0.37    | 0.86   | 0.40    |
|                   | $Sa_{avg2}(T)$ [g] | 0.21   | 0.27    | 0.38   | 0.29    | 0.54   | 0.28    | 0.78   | 0.36    |
|                   | $FIV3(T)$ [cm/s]   | 19.58  | 0.07    | 34.27  | 0.15    | 50.36  | 0.16    | 78.53  | 0.28    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.17   | 0.53    | 0.39   | 0.45    | 0.67   | 0.45    | 1.08   | 0.62    |
|                   | $Sa_{avg2}(T)$ [g] | 0.17   | 0.40    | 0.37   | 0.36    | 0.60   | 0.34    | 0.93   | 0.44    |
|                   | $FIV3(T)$ [cm/s]   | 15.39  | 0.19    | 33.70  | 0.16    | 53.35  | 0.23    | 85.22  | 0.29    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.15   | 0.55    | 0.35   | 0.45    | 0.54   | 0.46    | 0.77   | 0.46    |
|                   | $Sa_{avg2}(T)$ [g] | 0.14   | 0.45    | 0.31   | 0.30    | 0.46   | 0.39    | 0.74   | 0.44    |
|                   | $FIV3(T)$ [cm/s]   | 13.11  | 0.15    | 28.33  | 0.19    | 44.39  | 0.34    | 61.97  | 0.25    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.15   | 0.55    | 0.35   | 0.48    | 0.53   | 0.52    | 1.12   | 0.67    |
|                   | $Sa_{avg2}(T)$ [g] | 0.15   | 0.46    | 0.34   | 0.33    | 0.49   | 0.44    | 0.88   | 0.61    |
|                   | $FIV3(T)$ [cm/s]   | 14.43  | 0.24    | 29.69  | 0.19    | 44.72  | 0.21    | 85.02  | 0.39    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.16   | 0.52    | 0.41   | 0.37    | 0.60   | 0.44    | 0.80   | 0.44    |
|                   | $Sa_{avg2}(T)$ [g] | 0.15   | 0.42    | 0.35   | 0.35    | 0.55   | 0.36    | 0.74   | 0.38    |
|                   | $FIV3(T)$ [cm/s]   | 13.91  | 0.21    | 30.11  | 0.07    | 45.93  | 0.23    | 67.14  | 0.32    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 10.** Fragility function parameters for RC buildings in L'Aquila for T=0.33 using FEMAP695 set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.51   | 0.28    | 0.89   | 0.07    | 1.07   | 0.23    | 1.41   | 0.31    |
|                   | $Sa_{avg2}(T)$ [g] | 0.45   | 0.21    | 0.77   | 0.06    | 1.03   | 0.07    | 1.30   | 0.23    |
|                   | $FIV3(T)$ [cm/s]   | 51.99  | 0.25    | 90.75  | 0.24    | 118.25 | 0.12    | 167.29 | 0.07    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.49   | 0.25    | 0.89   | 0.07    | 1.11   | 0.23    | 1.57   | 0.28    |
|                   | $Sa_{avg2}(T)$ [g] | 0.45   | 0.21    | 0.76   | 0.06    | 1.06   | 0.06    | 1.33   | 0.17    |
|                   | $FIV3(T)$ [cm/s]   | 49.88  | 0.29    | 88.70  | 0.26    | 125.42 | 0.17    | 165.72 | 0.05    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.30   | 0.08    | 0.52   | 0.16    | 0.75   | 0.26    | 1.11   | 0.42    |
|                   | $Sa_{avg2}(T)$ [g] | 0.25   | 0.08    | 0.47   | 0.16    | 0.67   | 0.20    | 0.93   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 31.72  | 0.26    | 61.15  | 0.06    | 88.80  | 0.06    | 127.01 | 0.19    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.28   | 0.07    | 0.42   | 0.23    | 0.67   | 0.29    | 1.00   | 0.38    |
|                   | $Sa_{avg2}(T)$ [g] | 0.24   | 0.08    | 0.40   | 0.16    | 0.59   | 0.17    | 0.84   | 0.26    |
|                   | $FIV3(T)$ [cm/s]   | 27.35  | 0.25    | 51.07  | 0.15    | 75.20  | 0.14    | 111.23 | 0.14    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.21   | 0.21    | 0.43   | 0.28    | 0.67   | 0.30    | 1.17   | 0.46    |
|                   | $Sa_{avg2}(T)$ [g] | 0.19   | 0.14    | 0.39   | 0.22    | 0.61   | 0.27    | 0.98   | 0.50    |
|                   | $FIV3(T)$ [cm/s]   | 21.67  | 0.26    | 50.89  | 0.12    | 78.97  | 0.14    | 127.34 | 0.32    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.18   | 0.24    | 0.36   | 0.22    | 0.60   | 0.30    | 0.90   | 0.39    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.22    | 0.36   | 0.06    | 0.52   | 0.30    | 0.79   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 19.43  | 0.22    | 41.81  | 0.07    | 65.55  | 0.20    | 95.46  | 0.15    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.20   | 0.33    | 0.38   | 0.24    | 0.60   | 0.44    | 1.25   | 0.80    |
|                   | $Sa_{avg2}(T)$ [g] | 0.17   | 0.25    | 0.35   | 0.21    | 0.50   | 0.26    | 0.93   | 0.38    |
|                   | $FIV3(T)$ [cm/s]   | 21.05  | 0.20    | 43.57  | 0.07    | 67.93  | 0.29    | 114.47 | 0.40    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.18   | 0.26    | 0.38   | 0.28    | 0.62   | 0.38    | 0.89   | 0.41    |
|                   | $Sa_{avg2}(T)$ [g] | 0.17   | 0.21    | 0.38   | 0.05    | 0.58   | 0.25    | 0.79   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 21.05  | 0.20    | 46.93  | 0.17    | 71.49  | 0.20    | 104.71 | 0.26    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

**Table 11.** Fragility function parameters for RC buildings in L'Aquila for T=0.45 using FEMAP695 set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.46   | 0.31    | 0.79   | 0.21    | 1.03   | 0.25    | 1.37   | 0.33    |
|                   | $Sa_{avg2}(T)$ [g] | 0.39   | 0.30    | 0.66   | 0.02    | 0.90   | 0.06    | 1.22   | 0.15    |
|                   | $FIV3(T)$ [cm/s]   | 52.20  | 0.37    | 90.04  | 0.28    | 123.22 | 0.16    | 176.14 | 0.12    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.44   | 0.30    | 0.79   | 0.21    | 1.08   | 0.22    | 1.45   | 0.33    |
|                   | $Sa_{avg2}(T)$ [g] | 0.39   | 0.29    | 0.68   | 0.06    | 0.94   | 0.07    | 1.29   | 0.15    |
|                   | $FIV3(T)$ [cm/s]   | 50.51  | 0.35    | 91.55  | 0.28    | 122.67 | 0.19    | 183.21 | 0.17    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.26   | 0.25    | 0.52   | 0.16    | 0.73   | 0.21    | 1.02   | 0.37    |
|                   | $Sa_{avg2}(T)$ [g] | 0.22   | 0.08    | 0.43   | 0.14    | 0.64   | 0.07    | 0.89   | 0.20    |
|                   | $FIV3(T)$ [cm/s]   | 31.14  | 0.30    | 55.80  | 0.25    | 84.04  | 0.22    | 130.36 | 0.21    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.25   | 0.26    | 0.44   | 0.18    | 0.62   | 0.21    | 0.94   | 0.43    |
|                   | $Sa_{avg2}(T)$ [g] | 0.21   | 0.09    | 0.36   | 0.15    | 0.54   | 0.12    | 0.81   | 0.24    |
|                   | $FIV3(T)$ [cm/s]   | 28.69  | 0.25    | 51.44  | 0.23    | 77.71  | 0.23    | 113.97 | 0.22    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.18   | 0.06    | 0.41   | 0.18    | 0.62   | 0.32    | 1.10   | 0.53    |
|                   | $Sa_{avg2}(T)$ [g] | 0.17   | 0.18    | 0.37   | 0.18    | 0.56   | 0.20    | 0.89   | 0.45    |
|                   | $FIV3(T)$ [cm/s]   | 21.90  | 0.39    | 52.86  | 0.30    | 89.71  | 0.22    | 135.13 | 0.39    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.17   | 0.07    | 0.34   | 0.23    | 0.54   | 0.36    | 0.86   | 0.39    |
|                   | $Sa_{avg2}(T)$ [g] | 0.14   | 0.08    | 0.32   | 0.06    | 0.49   | 0.28    | 0.72   | 0.33    |
|                   | $FIV3(T)$ [cm/s]   | 18.72  | 0.34    | 43.40  | 0.16    | 66.36  | 0.21    | 102.56 | 0.22    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.18   | 0.25    | 0.34   | 0.28    | 0.57   | 0.25    | 1.08   | 0.48    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.23    | 0.33   | 0.17    | 0.46   | 0.06    | 0.95   | 0.43    |
|                   | $FIV3(T)$ [cm/s]   | 21.17  | 0.25    | 46.83  | 0.21    | 75.23  | 0.37    | 127.72 | 0.35    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.18   | 0.28    | 0.36   | 0.25    | 0.61   | 0.31    | 0.86   | 0.45    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.19    | 0.34   | 0.05    | 0.55   | 0.28    | 0.77   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 20.53  | 0.26    | 47.16  | 0.18    | 84.54  | 0.19    | 103.59 | 0.22    |

$\eta$ : median;  $\beta$ : dispersion.

*MSA*: multiple stripe analysis.

*CS*: conditional spectrum.

**Table 12.** Fragility function parameters for RC buildings in L’Aquila for T=0.7 using FEMAP695 set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.33   | 0.43    | 0.53   | 0.35    | 0.69   | 0.31    | 1.01   | 0.36    |
|                   | $Sa_{avg2}(T)$ [g] | 0.30   | 0.28    | 0.53   | 0.26    | 0.75   | 0.17    | 0.94   | 0.22    |
|                   | $FIV3(T)$ [cm/s]   | 47.57  | 0.43    | 86.66  | 0.36    | 108.18 | 0.35    | 168.65 | 0.20    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.33   | 0.43    | 0.51   | 0.37    | 0.73   | 0.27    | 1.07   | 0.31    |
|                   | $Sa_{avg2}(T)$ [g] | 0.29   | 0.27    | 0.54   | 0.25    | 0.81   | 0.13    | 1.05   | 0.12    |
|                   | $FIV3(T)$ [cm/s]   | 45.90  | 0.41    | 86.66  | 0.36    | 115.07 | 0.34    | 161.87 | 0.28    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.18   | 0.34    | 0.33   | 0.31    | 0.48   | 0.35    | 0.77   | 0.33    |
|                   | $Sa_{avg2}(T)$ [g] | 0.19   | 0.23    | 0.31   | 0.19    | 0.48   | 0.06    | 0.73   | 0.24    |
|                   | $FIV3(T)$ [cm/s]   | 28.88  | 0.32    | 49.95  | 0.31    | 75.06  | 0.34    | 124.49 | 0.16    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.16   | 0.31    | 0.28   | 0.26    | 0.45   | 0.24    | 0.67   | 0.28    |
|                   | $Sa_{avg2}(T)$ [g] | 0.16   | 0.22    | 0.27   | 0.21    | 0.40   | 0.15    | 0.61   | 0.18    |
|                   | $FIV3(T)$ [cm/s]   | 24.75  | 0.37    | 41.75  | 0.34    | 67.48  | 0.28    | 106.18 | 0.18    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.13   | 0.30    | 0.30   | 0.21    | 0.50   | 0.30    | 0.75   | 0.41    |
|                   | $Sa_{avg2}(T)$ [g] | 0.12   | 0.29    | 0.27   | 0.16    | 0.45   | 0.19    | 0.64   | 0.26    |
|                   | $FIV3(T)$ [cm/s]   | 19.36  | 0.52    | 44.05  | 0.36    | 75.15  | 0.23    | 119.93 | 0.34    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.12   | 0.31    | 0.28   | 0.19    | 0.42   | 0.25    | 0.60   | 0.31    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.24    | 0.24   | 0.19    | 0.39   | 0.18    | 0.53   | 0.15    |
|                   | $FIV3(T)$ [cm/s]   | 17.59  | 0.45    | 37.27  | 0.32    | 61.31  | 0.28    | 94.82  | 0.07    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.13   | 0.30    | 0.30   | 0.20    | 0.40   | 0.18    | 0.79   | 0.49    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.25    | 0.24   | 0.23    | 0.37   | 0.23    | 0.59   | 0.35    |
|                   | $FIV3(T)$ [cm/s]   | 18.54  | 0.36    | 37.84  | 0.32    | 58.41  | 0.35    | 123.61 | 0.48    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.13   | 0.31    | 0.30   | 0.16    | 0.45   | 0.28    | 0.67   | 0.40    |
|                   | $Sa_{avg2}(T)$ [g] | 0.11   | 0.24    | 0.26   | 0.19    | 0.42   | 0.17    | 0.55   | 0.24    |
|                   | $FIV3(T)$ [cm/s]   | 17.38  | 0.35    | 40.44  | 0.35    | 68.04  | 0.22    | 98.88  | 0.07    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

**Table 13.** Fragility function parameters for RC buildings in L'Aquila for T=1.5 using FEMAP695 set of records.

| Taxonomy          | IM                 | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|--------------------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |                    | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $Sa(T)$ [g]        | 0.14   | 0.50    | 0.24   | 0.48    | 0.28   | 0.41    | 0.46   | 0.39    |
|                   | $Sa_{avg2}(T)$ [g] | 0.13   | 0.41    | 0.24   | 0.35    | 0.29   | 0.36    | 0.43   | 0.28    |
|                   | $FIV3(T)$ [cm/s]   | 36.19  | 0.59    | 62.72  | 0.63    | 86.36  | 0.55    | 131.01 | 0.47    |
| CR-LFINF-CDM-7-H1 | $Sa(T)$ [g]        | 0.14   | 0.50    | 0.24   | 0.43    | 0.29   | 0.41    | 0.46   | 0.39    |
|                   | $Sa_{avg2}(T)$ [g] | 0.13   | 0.47    | 0.23   | 0.36    | 0.30   | 0.37    | 0.45   | 0.33    |
|                   | $FIV3(T)$ [cm/s]   | 35.45  | 0.53    | 62.72  | 0.63    | 86.36  | 0.55    | 124.60 | 0.50    |
| CR-LFINF-CDL-7-H2 | $Sa(T)$ [g]        | 0.08   | 0.51    | 0.15   | 0.41    | 0.20   | 0.40    | 0.34   | 0.29    |
|                   | $Sa_{avg2}(T)$ [g] | 0.08   | 0.38    | 0.14   | 0.37    | 0.21   | 0.33    | 0.34   | 0.20    |
|                   | $FIV3(T)$ [cm/s]   | 22.36  | 0.56    | 39.22  | 0.54    | 55.43  | 0.47    | 94.59  | 0.36    |
| CR-LFINF-CDM-7-H2 | $Sa(T)$ [g]        | 0.08   | 0.54    | 0.12   | 0.39    | 0.19   | 0.39    | 0.32   | 0.30    |
|                   | $Sa_{avg2}(T)$ [g] | 0.07   | 0.42    | 0.12   | 0.40    | 0.18   | 0.40    | 0.29   | 0.23    |
|                   | $FIV3(T)$ [cm/s]   | 21.39  | 0.57    | 33.84  | 0.64    | 48.80  | 0.61    | 78.78  | 0.62    |
| CR-LFINF-CDL-7-H3 | $Sa(T)$ [g]        | 0.05   | 0.59    | 0.13   | 0.52    | 0.23   | 0.46    | 0.37   | 0.42    |
|                   | $Sa_{avg2}(T)$ [g] | 0.06   | 0.47    | 0.12   | 0.34    | 0.21   | 0.40    | 0.39   | 0.50    |
|                   | $FIV3(T)$ [cm/s]   | 15.37  | 0.71    | 35.38  | 0.52    | 56.19  | 0.62    | 106.12 | 0.58    |
| CR-LFINF-CDM-7-H3 | $Sa(T)$ [g]        | 0.05   | 0.53    | 0.11   | 0.36    | 0.18   | 0.32    | 0.28   | 0.37    |
|                   | $Sa_{avg2}(T)$ [g] | 0.05   | 0.39    | 0.11   | 0.33    | 0.17   | 0.31    | 0.27   | 0.31    |
|                   | $FIV3(T)$ [cm/s]   | 13.49  | 0.65    | 29.23  | 0.54    | 44.89  | 0.45    | 72.43  | 0.34    |
| CR-LFINF-CDL-7-H4 | $Sa(T)$ [g]        | 0.05   | 0.47    | 0.10   | 0.45    | 0.17   | 0.43    | 0.39   | 0.48    |
|                   | $Sa_{avg2}(T)$ [g] | 0.05   | 0.36    | 0.11   | 0.38    | 0.17   | 0.41    | 0.36   | 0.52    |
|                   | $FIV3(T)$ [cm/s]   | 14.64  | 0.53    | 30.06  | 0.51    | 43.81  | 0.59    | 102.32 | 0.61    |
| CR-LFINF-CDM-7-H4 | $Sa(T)$ [g]        | 0.05   | 0.50    | 0.11   | 0.35    | 0.19   | 0.29    | 0.29   | 0.22    |
|                   | $Sa_{avg2}(T)$ [g] | 0.05   | 0.34    | 0.12   | 0.31    | 0.19   | 0.24    | 0.29   | 0.20    |
|                   | $FIV3(T)$ [cm/s]   | 13.80  | 0.56    | 32.04  | 0.48    | 48.99  | 0.43    | 73.58  | 0.33    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.

**Table 14.** Fragility function parameters for RC buildings in L'Aquila for  $PGA$  using FEMAP695 set of records.

| Taxonomy          | IM        | DS1    |         | DS2    |         | DS3    |         | DS4    |         |
|-------------------|-----------|--------|---------|--------|---------|--------|---------|--------|---------|
|                   |           | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ | $\eta$ | $\beta$ |
| CR-LFINF-CDL-7-H1 | $PGA$ [g] | 0.23   | 0.26    | 0.39   | 0.20    | 0.50   | 0.17    | 0.66   | 0.25    |
| CR-LFINF-CDM-7-H1 | $PGA$ [g] | 0.22   | 0.25    | 0.39   | 0.22    | 0.52   | 0.19    | 0.70   | 0.17    |
| CR-LFINF-CDL-7-H2 | $PGA$ [g] | 0.12   | 0.20    | 0.25   | 0.16    | 0.33   | 0.20    | 0.48   | 0.30    |
| CR-LFINF-CDM-7-H2 | $PGA$ [g] | 0.11   | 0.23    | 0.21   | 0.21    | 0.30   | 0.25    | 0.45   | 0.27    |
| CR-LFINF-CDL-7-H3 | $PGA$ [g] | 0.09   | 0.23    | 0.21   | 0.21    | 0.32   | 0.23    | 0.51   | 0.41    |
| CR-LFINF-CDM-7-H3 | $PGA$ [g] | 0.08   | 0.26    | 0.17   | 0.21    | 0.29   | 0.26    | 0.38   | 0.32    |
| CR-LFINF-CDL-7-H4 | $PGA$ [g] | 0.09   | 0.33    | 0.18   | 0.28    | 0.29   | 0.24    | 0.52   | 0.67    |
| CR-LFINF-CDM-7-H4 | $PGA$ [g] | 0.09   | 0.28    | 0.19   | 0.24    | 0.32   | 0.25    | 0.43   | 0.31    |

$\eta$ : median;  $\beta$ : dispersion.

MSA: multiple stripe analysis.